ISyE 3030 Project: Predictive Maintenance and Failure Classification in Manufacturing Systems
Team members' names: Phoenix Moore, Bill Nguyen, Sarah Wang, Ashley Lu, Cindy Le

I. Introduction

In modern manufacturing plants, machine failure can often create a decrease in operational efficiency, supply chain stagnation, and increase downtime, overall leading to a decrease in profit. Modern repair methods simply aim to fix the machine, but knowing when and how a machine fails would assist companies in preparing for these outages and preventing them from happening in the first place. By analyzing existing data on machine failures, industrial systems can not only optimize when and how resources flow through long operational lines to meet quotas, but also avoid failures and stops by changing factory conditions.

For this project, we hoped to identify what operational factors typically lead to machine failure. Through this, we hope to see whether certain factors impact a machine's success more than others and if we can create a model to predict failures before they occur.

II. Data Collection and Preprocessing

We used the AI4I 2020 Predictive Maintenance Dataset (Stephan, 2020), which is a publicly available dataset from the UCI Machine Learning Repository. The dataset contains 10,000 operational data metrics, including the machine's unique identifier (UID), machine failure, air/process temperature, standard deviation, and five independent failure modes. If at least one of the failure modes is true, the "machine failure" column is set to 1.

To prepare the dataset for statistical modeling, we used a data preprocessing pipeline using R's tidyverse package. We first validated the dataset by making sure the dimensions were correct and that there weren't any null or empty values. Next, we cleaned the data by standardizing the column names into an R-friendly format and dropping unneeded columns, specifically the UID and Product_ID columns. To accurately classify our models, we converted the product Type and the Machine_Failure outcome into categorical factors. Finally, we created a new predictive feature, Temperature Differential, by subtracting the air temperature from the process temperature to better capture the strain on the machines.

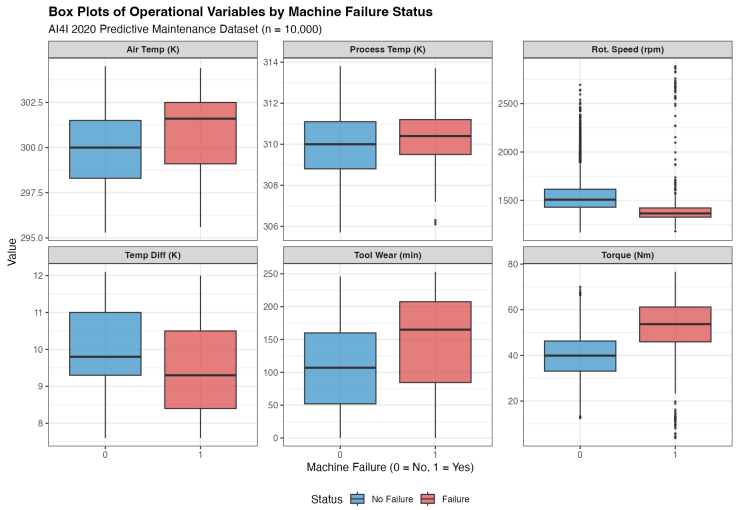
III. Descriptive Statistics

Descriptive Statistics - Operational Variables

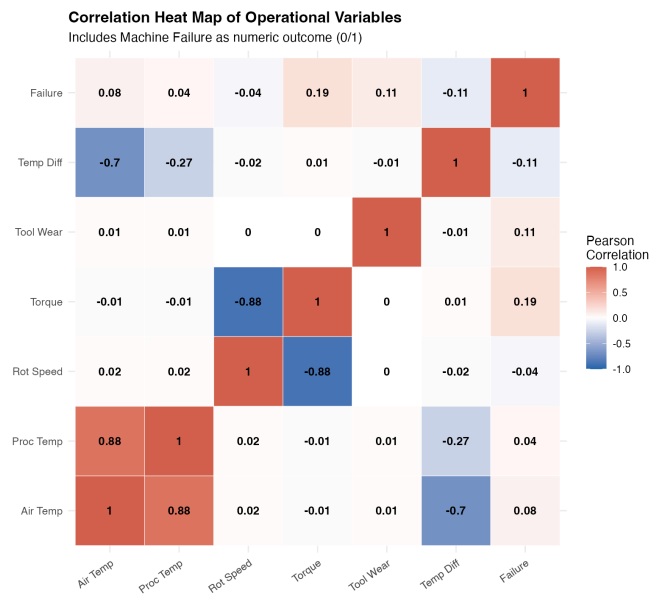
AI4I 2020 Predictive Maintenance Dataset | n = 10,000

Variable	Mean	Median	Std_Dev	Variance	Min	Max
Air Temperature (K)	300.005	300.1	2.000	4.001	295.3	304.5
Process Temperature (K)	310.006	310.1	1.484	2.201	305.7	313.8
Rotational Speed (rpm)	1538.776	1503.0	179.284	32142.787	1168.0	2886.0
Torque (Nm)	39.987	40.1	9.969	99.380	3.8	76.6
Tool Wear (min)	107.951	108.0	63.654	4051.850	0.0	253.0
Temp Differential (K)	10.001	9.8	1.001	1.002	7.6	12.1

Descriptive Statistics: This table shows the baseline operating conditions of the simulated factory floor. It shows the mean and spread of temperatures, torque, and rotational speed before filtering for failure events.



Tool A (Box Plot): To visually see the distribution of physical stress during the breakdowns, we generated a side-by-side box plot to compare each operational variable for either no failure or failure machines. As seen in the figure, machines that are in failure have a wider variance and a higher median torque. This suggests that extreme rotational resistance is a strong indicator of interruptions/shutdowns on the factory floor.

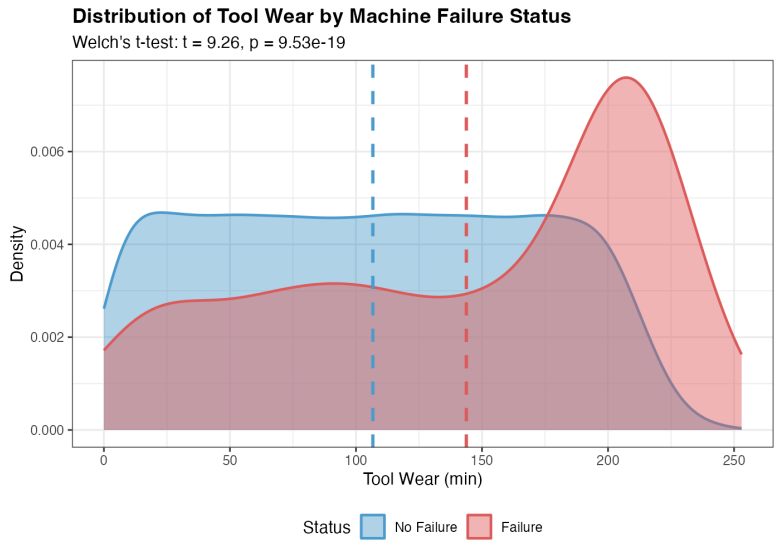


Tool B (Correlation Heatmap): We used a correlation matrix heatmap to evaluate the correlation between the continuous sensors. The matrix shows a strong inverse relationship between rotational speed and torque, as well as the expected positive correlation between air and process temperatures.

IV. Statistical Inference

Hypothesis: **Machines with higher wear on them are more likely to fail.**

- **Null Hypothesis (H_0):** There is no difference in machine wear between failed and operating machines.
- **Alternative Hypothesis (H_1):** There is a significant difference in machine wear between failed and operating machines.



We chose to do a **two-sample T-Test** because we are comparing the means of a continuous variable, machine wear, across two independent groups that are categorical, which are machine failure vs. non-failure.

Analysis: After performing the t-test, we received a p-value of approximately 0. This indicates that we will be rejecting our null hypothesis because our p-value of approximately 0 is less than our test statistic, which is alpha=0.05. From our findings in the t-test, we can say that we have statistically significant data to suggest that machines with higher wear on them are more likely to fail.

V. Regression Analysis

The Model: To model the relationship between operational variables and machine breakdown, we performed a Logistic Regression predicting the Machine Failure variable using Rotational Speed, Torque, and Tool Wear.

Logistic Regression — Coefficient & Odds Ratio Table

Outcome: Machine Failure (0/1) | Reference Type: H (High quality)

Predictor	Estimate	Std_Error	z_value	p_value	OR	CI_2.5pct	CI_97.5pct
(Intercept)	-37.0136	14.6652	-2.5239	1.161e-02	0.0000	0.0000	0.0003
Air_Temp_K	0.7700	0.0724	10.6352	2.045e-26	2.1597	1.8774	2.4939
Process_Temp_K	-0.7410	0.0964	-7.6851	1.529e-14	0.4766	0.3936	0.5745
Rotational_Speed_rpm	0.0115	0.0005	21.6705	3.893e-104	1.0116	1.0105	1.0127
Torque_Nm	0.2806	0.0115	24.4181	1.099e-131	1.3239	1.2950	1.3547
Tool_Wear_min	0.0133	0.0011	11.7081	1.158e-31	1.0134	1.0112	1.0157
Type_L	0.5958	0.2614	2.2794	2.264e-02	1.8145	1.1134	3.1149
Type_M	0.2774	0.2798	0.9915	3.215e-01	1.3197	0.7774	2.3388

Parameters & Accuracy: As shown in the summary, the model successfully converged. We evaluated the parameter significance by looking at p-values ($P > |z|$). The predictors for Tool Wear and Torque demonstrated high statistical significance. This validates our earlier tests.

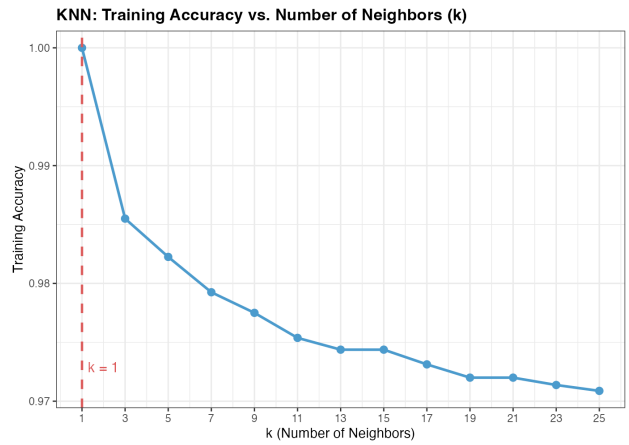
Implications: This establishes a strong baseline mathematically. It implies that while individual factors like torque contribute to failure, they can be combined into a linear equation to estimate the probability of a machine going down.

VI. Advanced Statistical Method

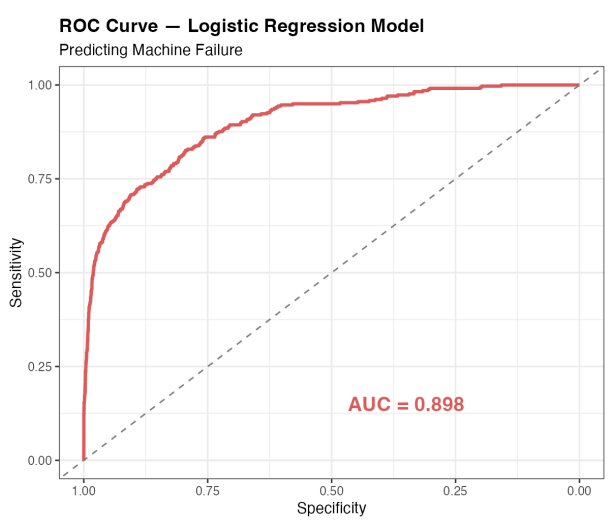
Model Comparison — Logistic Regression vs. KNN (Test Set)

80/20 train-test split | Positive class: Machine Failure = 1

Method	Accuracy	Sensitivity	Specificity	Balanced_Acc
Logistic Regression	0.9675	0.2055	0.9964	0.6009
KNN (k = 1)	0.9675	0.4658	0.9865	0.7261



K-Nearest Neighbors Model (top and bottom right): We tested our KNN model at k=1, and it achieved a high accuracy of 96.75%, which seems great from the outside, but that value is driven by the model's ability to classify the majority negative class, which was reflected in the high specificity of 98.65%. The sensitivity of our KNN model was much lower, at 46.58%, which means that our model was only identifying approximately 47% of the positive cases, suggesting that many positives are rare or are surrounded by negative observations. In comparison to our linear regression model, where both the KNN and linear regression models have the same accuracy of 96.75%, the KNN model outperforms the logistic regression model in detecting positive cases. Although our logistic regression model achieved slightly higher specificity compared to our KNN model, the linear regression model was far more biased towards the negative class, making our KNN model more balanced and a useful model in our statistical analysis.



Receiver Operating Characteristic (ROC) Curve: We performed a receiver operating characteristic model to further compare our KNN model to a linear baseline. To determine model adequacy, we used McFadden's pseudo-R^2. Our analysis from above is confirmed by the ROC curve, because the KNN model has a larger area under the curve compared to the logistic regression model. This graph helped us to demonstrate that the KNN analysis is significantly better at distinguishing between functional and failing machines, compared to the linear regression model, which makes our KNN model optimal in our statistical analysis.

VII. Conclusion

We utilized the AI41 2020 Predictive Maintenance Dataset to analyze factors associated with machine failure and to develop predictive models for classification. Using descriptive statistics such as boxplots and correlation heat maps, we identified significant differences between failed and non-failed machines, specifically in Torque and Tool Wear. The two-sample t-test provided strong statistical evidence that machines with greater wear are more likely to fail. Our regression analysis further supported these findings by demonstrating that Tool Wear and Torque were highly statistically significant predictors. This highlights that operational conditions can be used to estimate the probability of a machine failing. Moreover, the K-Nearest Neighbors model outperformed the logistic regression model in identifying positive failure cases. Since the logistic regression model was more biased towards the negative class, the KNN model proved to be more balanced and useful for our statistical analysis and predictive maintenance solutions. Accurate failure prediction is highly valuable because it can reduce/prevent downtime, lower maintenance costs, and improve efficiency through proactive maintenance scheduling.

However, there are limitations to this project, specifically with our dataset. Because our dataset was simulated to replicate real machine behaviors, it may not fully capture the complexity of actual industrial environments. Another limitation is that the failure class was relatively rare, creating a class imbalance that may affect model sensitivity. Future research may benefit from applying these methods to real-time industrial data, exploring more predictive models, and implementing balance techniques to improve the accuracy of failure detection.

Overall, we were able to reject our null hypothesis and find strong statistical evidence that operational conditions like torque and tool wear can be used to estimate the probability of a machine failing.

VIII. Data and Code Availability

All data preprocessing, statistical inference, and machine learning models were programmed in R. To view the complete, reproducible source code and the clean dataset used for this analysis, please visit our project repository on GitHub:
https://github.com/bilnguyyenn/predictive-maintenance-classification/tree/main

IX. References

Stephan, S. (2020). AI4I 2020 Predictive Maintenance Dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5HS5C.